

OpticalClaw: An Interactive Agentic Digital Twin for Optical Communication System Design and Optimization

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Abstract *OpticalClaw is a remote interactive agent for optical-link design. It embeds a high-fidelity waveform-level digital twin, enabling natural-language link construction, simulation, result analysis, and iterative parameter optimization in a closed loop. ©2026 The Author(s)*

Review

High-speed optical transmission systems are being pushed toward higher symbol rates, higher-order modulation formats, dense wavelength-division multiplexing (WDM), coherent detection, and digital signal processing (DSP)-intensive receivers [1–5]. These trends enlarge the design space of a transmission link: transmitter settings, fiber propagation, optical filtering, receiver architecture, and DSP parameters must be considered together. Laboratory experiments are required for final validation, while candidate schemes are usually examined first in waveform-level simulation environments before hardware implementation. Platforms such as VPItransmissionMaker and OptiSystem provide system-level environments for optical-link design and analysis[6, 7]. In practice, however, using such environments for link exploration still depends on simulator-specific operation skills and optical-domain knowledge: a user must convert a link idea into an executable topology, select analyzers, interpret waveforms, spectra, constellations, received powers, and bit error rate (BER)-related results, and then tune the system based on experience.

Recent progress in large language model (LLM) agents can reduce this coordination burden. ReAct-style agents combine reasoning with external actions, tool-use agents expose executable functions to language models, and feedback agents revise later decisions from previous results [8–11]. Agent runtimes such as OpenClaw and Model Context Protocol (MCP)-style interfaces further enable language models to connect with tools, resources, and reusable domain skills [12, 13]. Related work has also explored artificial intelligence (AI), machine learning (ML), digital twins, physical-layer-aware planning, and LLM-assisted automation for optical communication and optical networks [14–23]. For optical transmission design,

the key point is that language interaction alone is insufficient. The agent must be connected to an executable waveform-level simulator, be able to create valid optical-link projects, operate component models and parameters, run simulations, read numerical and visual outputs, and revise the design according to optical evidence.

We propose OpticalClaw, a remote interactive agentic digital twin designed for this setting. OpticalClaw embeds a high-fidelity optical transmission digital twin and exposes it to an agent layer through tool interfaces and domain skills. The demonstrated technologies include natural-language requirement understanding, component-grounded topology planning, JSON-based topology-plan generation, rule-based validation, typed canvas-graph execution, waveform-level simulation, multimodal result analysis, and iterative repair or parameter optimization.

Novelty

OpticalClaw introduces a constrained agentic design mechanism for pre-experimental optical-transmission studies. This design addresses two limitations of current optical-link exploration: conventional simulators require substantial manual expertise, while unconstrained LLM agents may generate invalid topologies or unsupported simulator operations. As shown in Fig. 1, physical optical experiments remain essential but require instruments, hands-on skills, and domain expertise. OpticalClaw targets the earlier design stage by translating a natural-language request into an executable and inspectable waveform-level digital-twin workflow. In the demonstrated implementation, a GPT-5.5-based LLM serves as the reasoning and planning core, while topology construction, validation, simulation, and result retrieval are performed through explicit tools and digital-twin interfaces. The agent does not issue unconstrained simulator commands. It converts the request into a struc-

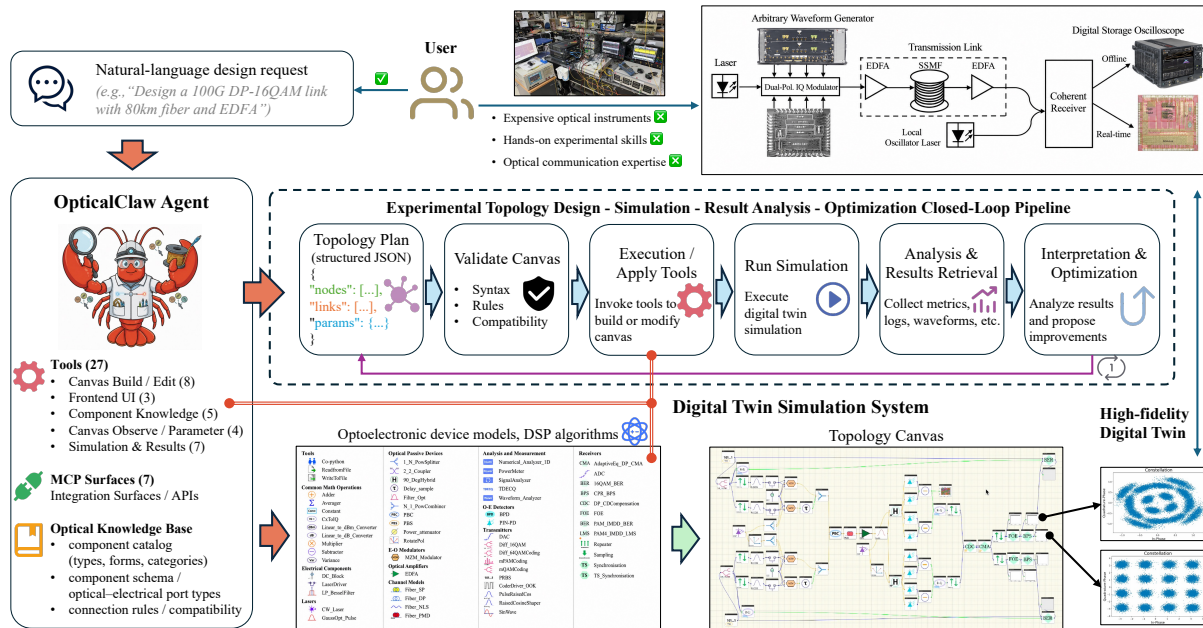


Fig. 1: Overall architecture of OpticalClaw for LLM-agent-driven optical-link design and waveform-level digital-twin simulation.

tured topology plan, validates the plan against component and connection knowledge, materializes it through typed graph-building actions, and then revises the design using simulation observations.

From an agent-design perspective, OpticalClaw separates planning, grounding, and execution. The LLM interprets the intent and generates a topology plan; the validator and executor ground the plan in component schemas, port types, connection rules, and simulator actions; and the interpretation module converts waveform-level observations into repair or tuning decisions. This mechanism is supported by three resources rather than fixed replayed templates. The tool registry provides 27 typed operations for canvas construction and editing, user-interface control, component-knowledge access, canvas observation, parameter inspection, simulation control, and result analysis. The MCP surfaces provide standardized integration between the agent, simulation runtime, client workspace, telemetry, and component-knowledge services. The optical knowledge base stores component catalogs, optical and electrical port definitions, connection rules, compatibility constraints, and topology design knowledge.

The closed-loop revision step distinguishes OpticalClaw from template-based construction. The topology plan records nodes, links, and parameters as an explicit intermediate representation between language and simulator execution. After validation, the executor builds or modifies the topology canvas and launches the digital-twin simulation. The underlying digital twin integrates optoelectronic device models, optical amplifiers, fiber-channel models, receiver front-end models, DSP algorithms, and analysis modules. The fiber-

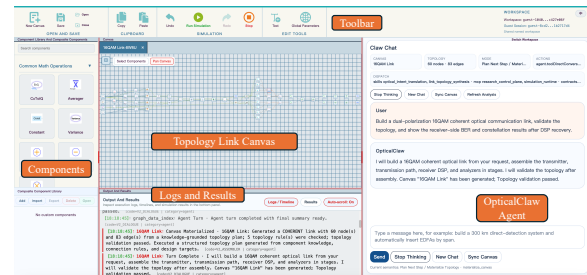


Fig. 2: OpticalClaw platform interface.

channel models capture chromatic dispersion and fiber nonlinearity through numerical solution of the nonlinear Schrodinger equation (NLSE). After each run, OpticalClaw retrieves waveforms, spectra, constellations, BER-related metrics, logs, and tables. These observations are interpreted to identify missing components, invalid connections, degraded signal quality, or suboptimal parameters, and the resulting repair or tuning actions are fed back to the next topology revision. Thus, optimization is driven by waveform-level simulation evidence rather than manual plot inspection alone. The platform interface in Fig. 2 exposes the topology canvas, agent dialogue, execution logs, and simulation results, making the workflow inspectable during live interaction. Representative interaction examples are shown in Fig. 3.

ECOC Relevance

OpticalClaw addresses ECOC topics including AI for optics, digital twins for optical systems and sub-systems, autonomous control, telemetry-based management, and AI- and ML-based optimization. For ECOC attendees working on optical transmission, DSP-intensive receivers, system-level simulation, and optical-network automation, the demo provides an inspectable example of how language

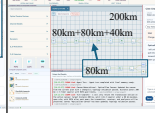
	Coherent DP-16QAM Link Construction	IM/DD Direct-Detection Link Construction	Simulation Execution	Result Analysis	Link-Length Reconfiguration
Chat	User: Build a dual-polarization 16QAM coherent optical communication link, validate the topology, and prepare receiver-side BER, constellation, and waveform analysis. OpticalClaw: I will build a 16QAM coherent optical link from your request, assemble the transmitter, transmission path, receiver DSP, and I will validate the topology after assembly. Canvas "16QAM Link" has been generated; Topology validation passed.	User: Build an OOK intensity-modulation direct-detection optical link, validate the topology, and prepare waveform analysis before and after coding as well as at the receiver. OpticalClaw: Canvas "graph data index" has been generated; Topology validation passed. - refresh analysis: refreshed 0 analysis results. Latest simulation ID: opticalclaw-v2-session-d82303db.	User: Run the simulation. OpticalClaw: Validate canvas: passed run simulation: started simulation bd03d14e13464231833fbd82117277b run simulation: simulation completed refresh analysis: refreshed 9 analysis results. Read 9 analysis results, analyzers include numerical analyzer 1d signal analyzer. Key metrics: BER=0, numerical mean=0.	User: Run the simulation again and compare the new results with the previous run. Focus on BER, constellation quality, waveform distortion, and whether the longer fiber link changes the dominant impairment. OpticalClaw: In progress • Processing • 35% I will validate "graph data KKU8", run the simulation, and refresh the analysis results after it completes.	User: Change the transmission distance to 200 km. OpticalClaw: I will only retune the transmission section on the active canvas: target distance 200 km, split into 3 span(s) under a 80 km maximum-span rule, with EDFA after each span; the transmitter, receiver, and analyzers will be preserved. Canvas "OpticalClaw Canvas" has been updated; Topology validation passed.
Action	The agent synthesizes the transmitter, transmission path, coherent receiver DSP, and analysis modules; the topology is materialized on the canvas and validated successfully.	The agent assembles the laser, modulator, fiber channel, photodetector, electrical receiver, and analysis modules for direct detection.	The agent executes the standard simulation chain: validate_canvas → run_simulation → refresh_analysis.	The agent retrieves analyzer outputs, simulation logs, BER-related metrics, constellation diagrams, and waveform results for interpretation.	The agent updates the fiber-span configuration, inserts EDFA-assisted spans when required, preserves the transmitter and receiver, and revalidates the modified topology.
Platform Evidence	 Generated DP-16QAM coherent-link canvas with validation-passed message and receiver-side analyzers.	 Generated IM/DD topology with direct-detection receiver and waveform/eye/BER analyzers.	 Runtime logs showing validation passed, simulation started, simulation completed, and analysis refreshed.	 Analysis window showing a 4x4 16-QAM constellation and retrieved simulation metrics.	 Updated transmission section showing 200km distance split into multiple spans with EDFA blocks.

Fig. 3: Representative OpticalClaw demonstration scenarios and platform evidence.

agents can be coupled to physically meaningful waveform-level simulations. The main value of the demo lies in the live integration of agentic planning, optical-domain constraints, executable topology construction, simulation feedback, and revision based on optical observables. During the demo, attendees can submit their own link-design requests, inspect the generated topology, validation logs, tool actions, simulation results, and revision behavior. This allows the community to discuss how future optical-system design workflows may use constrained language agents not only to generate candidate links, but also to validate, simulate, diagnose, and retune them using simulator evidence.

Demo Implementation

(i) Objectives and configuration. The objective is to demonstrate the closed-loop workflow in Fig. 1. OpticalClaw converts a natural-language request into a structured topology plan in JSON, validates the plan, invokes typed tools to build the topology canvas, runs waveform-level digital-twin simulation, retrieves analysis outputs, and then tunes parameters or repairs the topology when needed. The remote platform contains the agent service, optical knowledge base, MCP interfaces, validator, executor, embedded optical transmission digital twin, and web dashboard. As shown in Fig. 2, the dashboard exposes the component library, topology canvas, execution logs, simulation results, and agent dialogue. Representative platform interactions are summarized in Fig. 3. During the demo, attendees may either follow these examples or submit their own link-design requests.

(ii) Physical setup and equipment. The demonstration will run through a remote OpticalClaw platform. The agent, knowledge base, validation engine, simulation engine, and analysis dashboard are hosted on a laboratory server or cloud workstation, while the on-site presenter operates the system through a web workspace. At

the ECOC booth, a presenter laptop connected to the provided monitor and wireless local area network (WLAN) is used to access the platform. A backup mobile hotspot and short prerecorded runs will be prepared to improve demonstration resilience.

(iii) Presentation to attendees. The presenter will explain the OpticalClaw agent architecture and technical workflow in Fig. 1, introduce the front-end workspace regions and interaction process shown in Fig. 2, and then demonstrate examples similar to Fig. 3. These cases cover coherent dual-polarization 16-ary quadrature amplitude modulation (DP-16QAM) and intensity modulation/direct detection (IM/DD) link construction, simulation execution, result analysis, and link-length reconfiguration with erbium-doped fiber amplifier (EDFA)-assisted spans. For example, when the transmission distance is changed to 200 km, OpticalClaw can update the transmission section, split the link into EDFA-assisted spans under a maximum-span rule, preserve the transmitter, receiver, and analyzers, revalidate the canvas, rerun the simulation, and compare the new results with the previous run.

(iv) Attendee interaction. Attendees will be invited to submit their own natural-language requirements. They can ask OpticalClaw to construct a coherent or direct-detection link, run a simulation, inspect BER/constellation/waveform/spectrum results, modify the link length, add EDFA-assisted spans when needed, compare parameter settings, or repair an incomplete design. The interface exposes the mapping from text to components and connections, validation messages, executed tool actions, analysis outputs, and waveform-level feedback used for revision. This allows attendees to test not only whether a topology can be generated, but also how the agent reacts when simulation evidence suggests that the design should be repaired or retuned.

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